

View results

Respondent

48

Anonymous

300:59

Time to complete

A. General information about the respondent

1. * Country

Indonesia



2. * Name of organization(s) or entity(s) responsible for the preparation of the report

National Research and Innovation Agency

3. * Website of organization(s) or entity(s) responsible for the preparation of the report

<https://brin.go.id/en>

We'd love your feedback!



We have just two questions for you.

4. * Email address of organization(s) or entity(s) responsible for the preparation of the report

5. Phone number of organization(s) or entity(s) responsible for the preparation of the report

6. Please describe the role/mandate of your organization

National Research and Innovation Agency (BRIN) is tasked with monitoring, controlling, and evaluating the performance of duties and functions of Regional Research and Innovation Agencies (BRIDA) pursuant to the provisions of laws and regulations. Article 3 of Presidential Regulation No. 78/2021 concerning BRIN which was signed by President Jokowi on 24 August 2021 stipulates that BRIN has the task of assisting the President in performing government duties in the fields of research, development, studies, and application as well as invention and innovation, organization of nuclear energy and outer space operations in an integrated manner nationally.

7. * Full name of officially designated contact person(s) who completed this survey

Hendro Subagyo

8. * Position of officially designated contact person(s) who completed this survey

Director of Repository Multimedia and Scientific Publishing

9. * Email address of officially designated contact person(s) who completed this survey

10. Full Name(s) of designated official(s) certifying the report

Hendro Subagyo

11. Brief description of the consultation process established for the preparation of the report

First of all, we invite stakeholders that related with the implementation recommendation on the kick off meeting for the preparation report including non governmental organization, scientific community, and media. furthermore, we conduct the meeting through online platform and having the explanation briefly regarding the objective and position of the Survey. we clustered the participant into three categories (regulator, beneficiary, and facilitator). During one week, every participant/entities fill up the (online) document survey. After the document, we verify and confirm to the participant to get the clarity of their response if needed.

12. * Name of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

Good Science Indonesia (scientific community); The Conversation Indonesia; National Library Indonesia, LPDP (Indonesia Research Funding Agency), The Conversation Indonesia (Media), Wikimedia Indonesia; Creative Commons Indonesia

13. * Website of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey
(Please use commas to separate multiple links, as: link A, link B)

<https://theconversation.com/id>, <https://www.perpusnas.go.id/>,
<https://creativecommons.or.id/>, <https://wikimedia.or.id/>, <https://sisj-indonesia.org/>,
<https://lpdp.kemenkeu.go.id/>

14. * Email address of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

15. Sector of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

- ☒ Public
- ☒ Private
- ☒ Civil Society
- ☐ Other

B. QUESTIONNAIRE FOR MEMBER STATES' REPORTING

ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION ON OPEN
SCIENCE

Part 1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE,

ASSOCIATED BENEFITS AND CHALLENGES, AS WELL AS DIVERSE PATHS TO
OPEN SCIENCE

16. 1.1 Has the 2021 UNESCO Recommendation on Open Science been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

- ☒ Yes
- ☐ No

17. 1.1 cont. **If yes**, please indicate which ministries and or national authorities/entities have been involved in the promotion of the 2021 Recommendation.

Ministry of Higher Education and Science and Technology and BRIN. Both Institution play significant role to advancing 2021 recommendation

18. 1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

☒ Yes

☐ No

19. 1.2 cont. **If yes**, which language(s)?

Bahasa

20. 1.3 Have there been awareness raising activities on open science, including all its key elements[i], associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? (ref.: (i) 16.f, 16.g, 16.h)

[i] Please refer to the elements of open science defined in the 2021 Recommendation, namely : open scientific knowledge (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), open science infrastructures, open engagement of societal actors and open dialogue with other knowledge systems.

☒ Yes

☐ No

21. 1.3 cont. **If yes**, please provide details on the activities organized or foreseen and links as relevant.

Indonesia has demonstrated involvement in open science funding discussions at regional and international levels, although there is no consistency on follow-up. There are challenges in implementation due to the still dominant use of citation-based metrics in research assessment in national regulations, while UNESCO recommendations encourage broader open science practices such as data sharing, collaboration and transparency. Open science has several benefit such as increasing research efficiency, encourage innovation, and strengthen collaboration between researchers. However, without clear regulatory and funding mechanisms, this practice will be difficult to develop. Example : 1.

<https://jurnal.ugm.ac.id/bip/article/view/17054> 2. <https://www.csis.or.id/publication/a-case-for-open-science-in-indonesia-advancing-digital-literacy-scientific-infrastructure-and-leadership/> 3. RINarxiv preprint service 4. Open Science Meeting, Open Science Webinar 5. Blogpost, Science Watch Dog,

22. 1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open science[i], in publicly funded research? (ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)

[i] The 2021 Recommendation outlines a set of shared values of open science stemming from the rights-based, ethical, epistemological, economic, legal, political, social, multi-stakeholder and technological implications of opening science to society and broadening the principles of openness to the whole cycle of scientific research. It also provides a set of guiding principles, as a framework for enabling conditions and practices, which uphold the values of open science and can realize the ideals of open science.

Please see the values of open science, namely quality and integrity, collective benefit, equity and fairness, and diversity and inclusiveness.

Please see the guiding principles for open science ,namely transparency, scrutiny, critique and reproducibility; equality of opportunities; responsibility, respect and accountability; collaboration, participation and inclusion; flexibility and sustainability.

☒ Yes

☐ No

23. 1.4 cont. If **yes**, please indicate the actions that are taken to incorporate the specific values and principles. For each case, please provide details and links as relevant.

In several Collaborative Research and Innovation Funding for Advanced Indonesia (RIIM) schemes, organized by BRIN and LPDP, the implementation of open science has been encouraged. Link : [https://pendanaan-risnov.brin.go.id/program/eyJpdil6lldRSk9KdmthOW1YdkxIN1o3RjlzR3c9PSIsInZhbHVlIjoIS2p5VDBrVnZWdUlwTzk4Rkdicmpydz09IiwibWFjIjoieYzA1NTYwMWQzNmI1NWQwNDk2MjI1MlWl0ZmZjNjllYzRIYWQwZmQ2ZmI1M2RmNDBjODUzZDRkNmZlZWEzYjg3OSIsInRhZyl6IiJ9](https://pendanaan-risnov.brin.go.id/program/eyJpdil6lldRSk9KdmthOW1YdkxIN1o3RjlzR3c9PSIsInZhbHVlIjoIS2p5VDBrVnZWdUlwTzk4Rkdicmpydz09IiwibWFjIjoieYzA1NTYwMWQzNmI1NWQwNDk2MjI1MlWl0ZmZjNjllYzRIYWQwZmQ2ZmI1M2RmNDBjODUzZDRkNmZlZWEzYjg3OSIsInRhZyl6IiJ9;); Wikimedia Indonesia, through the Wikidata Indonesia program, organizes training and outreach that introduces the principles of Open Linked Data, open data sources and Wikidata. Documentation available at https://id.wikimedia.org/wiki/Wikidata_2024/Dokumentasi_aktivitas; However, it must be acknowledged that no concrete program has been planned or implemented by stakeholders with full awareness to fulfill UNESCO's recommendations. Here the difficulty arises, when someone takes the initiative, there will always be a response that so far various efforts have been made in line with UNESCO recommendations. Consolidation efforts are still needed to equalize perceptions and unite concrete steps for implementing Open Science in Indonesia.

Part 2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

24. 2.1 Does your country have a national policy[i], strategy or plan of action on science, technology and innovation (STI)?

[i] National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision-making processes in the area of STI on the national level.

☒ Yes

☐ No

25. 2.1 cont. **If yes**

- please share a machine-readable PDF file of the policy using this link: <https://tinyurl.com/27p5zct8> designating openscience@unesco.org as the recipient, and provide the following information: *Title of the policy, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, and links to the webpage if available.*

In the comment box, please indicate the question number for this attachment / document.

After uploading, please return to this page to continue the survey.

26. 2.1 cont. **If yes:**

- does it include any of the following open science elements:

- ☒ open access to scientific knowledge
- ☒ open science infrastructure
- ☒ open engagement of societal actors
- ☐ open dialogue with other knowledge systems

27. 2.2 In your country, is there a policy, or a set of policies and/or legal framework(s)[i] at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? (ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)

[i] These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.

☒ Yes

☐ Partly

☐ No

28. 2.2 cont. **If yes or partly**, please share a machine-readable PDF file(s) of the policy(ies) using this link: <https://tinyurl.com/27p5zct8> designating openscience@unesco.org as the recipient, and provide the following information for each: *Title of the policy or legal instrument, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, elements of open science that are addressed, and links to the webpage(s) if available.*

29. 2.3 In your country, are there specific policy instruments[i] that aim to promote open science?

[i] Policy instruments include programmes, methods and mechanisms of technical and operational nature, required to solve the issues set by the policy, with focus on the target beneficiaries, resources, indicators, and set of goals for delivering products (short term), results (medium-term), and impacts (long term).

- ☒ Yes
- ☐ Under development
- ☐ No

30. 2.3 cont. **If yes or under development**, please provide details and links as relevant, including for each instrument: *Title of the instrument, its objective, the authority in charge, allocated budget, elements of open science included.*

Majority of policy instruments are still focused on aspects of open access publications, especially those related to the management and development of open access journals. Although this is a positive step in supporting the dissemination of scientific knowledge, the scope of this policy is still limited and does not fully reflect the full spectrum of open science practices recommended by UNESCO.

31. 2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? (ref.: (ii) 17.a, 17.b, 17.c, 17.f, 17.h)

- ☒ Yes
- ☐ Under development
- ☐ No

32. 2.4 cont. **If yes or under development**, please provide details and links as relevant, including the name of ministries and national authorities/entities, as well as affiliated organizations that are involved in the development and implementation of the plan, strategy or the roadmap.

Although there are several programs that have been implemented currently, such as ELSA and RIN (open scientific services for public provided by National Research and Innovation Agency) for storing and sharing research data, there is still no visible development of new programs that comprehensively lead to the implementation of the UNESCO Recommendations on Open Science. Existing programs, although making positive contributions, are still limited in scope and have not fully adopted the open science principles recommended by UNESCO.

33. 2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at **institutional** level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? (ref.: (ii) 17.d, 17.e, 17.g)

- ☒ Yes
- ☐ Under development
- ☐ No

34. 2.5 cont. **If yes**, please provide more information, including the number of policies, name of the institute(s) that has/have developed the policy(s) and elements of open science included in the policy (please consider policies and strategies addressing all elements of open science, including citizen and participatory science, or co-production of knowledge with multiple actors).

To view the elements of open science,

see: <https://unesdoc.unesco.org/ark:/48223/pf0000379949/PDF/379949eng.pdf.multi.nameddest=20/p.nameddest=20/.page=9>

BRIN has an open platform policy that opens up access to infrastructure, funding and resources to carry out research and collaboration between various parties from universities, BRIN and industry.

35. 2.6 In your country, are there specific funding mechanisms[i] for open science at the national level? (ref.: (ii) 17.j, (iii) 18.j, (iv) 19.c)

[i] A funding mechanism refers to a policy instrument or process through which financial resources are allocated or provided for a specific purpose. Some examples are block funding, research grants, support for technological poles and centres of excellence, support for science parks and innovation centres, Infrastructure grants (for research facilities, labs, instruments), communication and outreach funds, innovation funds, loans and tax credits scholarships, studentships, fellowships, trust funds, sectoral funds.

- ☐ Yes
- ☒ Under development
- ☐ No

36. 2.6 cont. **If yes or under development**, please provide details and links as relevant.

37. 2.7 1.1 In your country, have open science practices[i] been integrated in existing research funding mechanisms?

[i] Examples can include sharing open scientific knowledge (publications, research data, educational resources, software and hardware) with other scientists and with the public, sharing or using open infrastructures, collaboration with other researchers, with national and local societal actors such as policy makers, or sectors such as industry, agriculture, and health, and collaboration and open dialogue with indigenous peoples and local communities.

☒ Yes

☐ No

38. 2.7 cont. **If yes**, please provide details and links as relevant.

Creative Commons Indonesia (CCID) in implementing the Open Learning Resources (OER) project in (2024-2025) received funding support from the Wikimedia Foundation. This SPT project began by holding an SPT collaboration forum in September 2024. At the forum, various types of SPT collaboration actions were agreed, namely: • SPT online campaign which is participated in by almost all SPT forum participants via Instagram which will be held from October 2024 – April 2025. For the campaign theme in March, we will carry the theme of open science. (online campaign link) • Teaching material illustration contest (KIBA) held in October - November 2024 with CCID and the Gimpscape community and supported by Seameo Seamolec, an Indonesian learning teacher community. From this contest, 257 illustrations of teaching materials were obtained from 16 participants. • The talk show and editathon for Science Teaching Materials on the Internet which will be held on 5-6 December 2024 is a collaboration between BBGP West Java and Wikimedia Indonesia. This event was attended by 36 teachers with a science education background at secondary school level. All participants successfully uploaded at least one work to Commons and curated science illustrations available on Wikimedia Commons. These science illustrations, which can be used as teaching materials, are classified as OER and are now available for other science educators in Indonesia to use. (The results of curation of science teaching materials by educators can be accessed here) • Open learning source online training in December 2024 - May 31 2025. This training is supported by LMS from SEAMEO SEAMOLEC with a total of 497 participants who are educators.

39. 2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (ref.: (ii) 17.i)

☐ Yes

☒ No

40. 2.8 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

The implementation regulation is under discussion

41. 2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy (ref.: (ii) 17.j, 23.c)

☐ Yes

☒ Under development

☐ No

42. 2.9 cont. **If yes or under development**, please provide details and links as relevant.

The Implementation regulation regarding the monitoring and evaluation is under discussion and harmonization with others related-regulation

Part 3. INVESTING IN
INFRASTRUCTURES AND S

43. 3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.a)

The official data regarding the percentage of GDP dedicated to research and development expenditure : <https://penerbit.brin.go.id/others/catalog/book/1621> ©2024 Badan Riset dan Inovasi Nasional Direktorat Pengukuran Indikator Riset, Teknologi, dan Inovasi Jakarta: Penerbit BRIN, 2024. x+74 halaman, 17,6 x 25 cm ISSN 2961-9831

44. 3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.b)

Indonesia telecommunication statistic year of 2023,
<https://www.bps.go.id/id/publication/2024/08/30/f4b846f397ea452bdc2178b3/statistik-telekomunikasi-indonesia-2023.html>

45. 3.3 Are there national research and education networks (NRENs)[i] active in the country? (ref.: (iii) 18.c)

[i] A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.

☒ Yes

☐ No

46. 3.3 cont. **If yes**, please provide more information and links as relevant, and indicate to which open science actors in your country those networks are accessible.

The Indonesian Research and Education Networks (IDREN) is a special network infrastructure that connects research and education institutions and communities in Indonesia. This network is dedicated to the data communications connectivity needed to enable a community of researchers, academics and developers to collaborate together. The IDREN network infrastructure was realized thanks to support and collaboration between the telecommunications industry community (network operators, internet service providers) which manages connectivity up to layer two (physical) and the research and education community which manages connectivity at layer three (network), as well as support for connectivity to global REN. Here is the link : <https://idren.id/>

47. 3.4 Are there national or institutional open science infrastructures[i] in your country? (ref.: (iii) 18.d, 18.e, 18.k, 18.l, 18.m)

[i] Some examples of open science infrastructures include major shared scientific equipment or sets of instruments, open computational and data manipulation service infrastructures that enable collaborative and multidisciplinary data analysis, open science platforms and repositories for publications, research data and source codes, archives, open bibliometrics and scientometrics systems for assessing and analysing scientific domains, open laboratories, software forges and virtual research environments, open innovation testbeds, incubators, science museums, science parks and infrastructure for non-digital materials.

- ☒ Yes
- ☐ Under development
- ☐ No

48. 3.4 cont. **If yes or under development**, please indicate which type(s) of infrastructure:

- ☒ federated information technology infrastructure for open science, including high-performance computing, cloud computing and data storage
- ☒ community managed infrastructures, protocols and standards, for example those that support bibliodiversity and engagement with society
- ☒ platforms for exchanges and co-creation of knowledge between scientists and society
- ☒ community-based monitoring and information systems to complement national, regional and global data and information systems

49. 3.4 cont. **If yes or under development**, for each infrastructure, please provide links as relevant and more information with regard to their compliance with the 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility and Ethics) principles. Please indicate if the infrastructure is non-commercial.

<https://elsa.brin.go.id/subkategori/index/BRIN%20HPC/18>; <https://rin.brin.go.id/>;
<https://creativecommons.or.id/2024/12/pelatihan-daring-sumber-pembelajaran-terbuka-2025-sinergi-untuk-berbagi-pengetahuan.html>

50. 3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (ref.: (iii) 18.e, 18.f)

☒ Yes

☐ No

51. 3.5. cont. **If yes**, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open science, stage of research, accessibility to all the relevant users in the region or internationally).

<https://elsa.brin.go.id/subkategori/index/BRIN%20HPC/18>

52. 3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? (ref.: (iii) 18.g)

☒ Yes

☐ No

53. 3.6 cont. **If yes**, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open science, stage of research, accessibility to all the relevant users in the region or internationally).

<https://international-relations.web.cern.ch/stakeholder-relations/states/indonesia>; ALICE in the area of grid computing.

Part 4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY AND CAPACITY BUILDING FOR OPEN SCIENCE

54. 4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? (ref.: (iv) 19.a, 19.c)

☐ Yes

☒ No

55. 4.1 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

There is no systematic capacity building program for policy makers. This is caused by limited understanding by policy makers regarding the concept and benefits of open science, as well as a lack of awareness of the importance of integrating open science principles into national Science Technology and Innovation policies. This limited insight is reflected in the lack of policy initiatives that explicitly support the implementation of open science practices at various levels.

56. 4.2 Have capacity building programmes for policy makers on how to integrate core values and principles of open science in STI policies and strategies taken place in your country?

☐ Yes

☒ No

57. 4.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

There is no systematic capacity building program for policy makers. This is caused by limited understanding by policy makers regarding the concept and benefits of open science, as well as a lack of awareness of the importance of integrating open science principles into national Science Technology and Innovation policies. This limited insight is reflected in the lack of policy initiatives that explicitly support the implementation of open science practices at various levels.

58. 4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? (ref.: (iv) 19.b)

☐ Yes

☒ No

59. 4.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Although the WIBISANA initiative of the Indonesian Open Science Community was launched five years ago and provides a comprehensive learning platform to introduce open science practices, to date there has been no systematic effort at the institutional or national level to integrate and develop open science components into the higher education curriculum. This lack of formal integration indicates a significant gap between the availability of learning resources and their implementation in the higher education system.

60. 4.4 Are open educational resources[i] as defined in the 2019 Recommendation on Open Educational Resources, used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? (ref.: (iv) 19.d)

[i] Open Educational Resources are learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others (2019 Recommendation on Open Educational Resources)

☒ Yes

☐ Partly

☐ No

61. 4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? (ref.: (iv) 19.e)

☐ Yes

☒ No

62. 4.5 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Although the WIBISANA initiative of the Indonesian Open Science Community was launched five years ago and provides a comprehensive learning platform to introduce open science practices, to date there has been no systematic effort at the institutional or national level to integrate and develop open science components into the higher education curriculum. This lack of formal integration indicates a significant gap between the availability of learning resources and their implementation in the higher education system.

Part 5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

63. 5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? (ref.: (v) 20.b, 20c, 20i)

☐ Yes

☒ No

64. 5.1 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

The majority, if not all, of the assessment and evaluation systems currently implemented still rely heavily on mainstream metrics such as the number of citations and journal impact factors. This approach focusing on quantitative metrics represents a significant gap with UNESCO recommendations, which encourage the use of more comprehensive and diverse assessment indicators. Current evaluation systems tend to ignore important aspects of open science practices such as sharing research data, cross-disciplinary collaboration and engagement with society, which are key elements in UNESCO's recommendations on open science

65. 5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref.: (v) 20.a, 20.c, 20.d)

☐ Yes

☒ No

66. 5.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Although BRIN has implemented several important programs such as ELSA and RIN (National Scientific Repository), open science initiatives at the government level are still very limited. The Ministry of Education and Technology, as one of the main stakeholders in the national research ecosystem, has not initiated significant programs in this context. What is more concerning is that most existing activities are still focused on downstream aspects of the research cycle, especially on open access publications, while other important aspects of open science such as data sharing, open methodology and collaboration have not received sufficient attention.

67. 5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? (ref.: (v) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20h)

☐ Yes

☒ No

68. 5.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

The Indonesian government has not developed a comprehensive reward system for open science initiatives. Although there are several programs supporting open science practices, the majority of existing initiatives are still limited to downstream aspects of the research process, particularly in the form of open access publications. Even in the context of these publications, the focus of evaluation and awards is more on citation-based measurements than on the quality of the content or substantive contribution of the research. This reflects the gap between prevailing evaluation practices and the spirit of open science which emphasizes transparency, accessibility and social impact of scientific research.

69. 5.4 Have there been any unintended negative consequences[i] of open science practices reported in your country? (ref.: (v) 20)

[i] For example, predatory behaviours, data migration, exploitation and privatization of research data, increased costs for scientists and high article processing charges associated with certain business models in scientific publishing that may be causes of inequality for the scientific communities around the world and, in some cases, the loss of intellectual property and knowledge.

☒ Yes

☐ No

70. 5.4 cont. **If yes**, please provide more information and specify if there is a strategy or plan to preventing and mitigating such consequences.

Until now, several concerns that often arise in discussions about open science have not been supported by strong arguments and sufficient empirical evidence. These concerns include several main aspects: First, there is a fear that when research results are freely disclosed, the work will be vulnerable to acts of plagiarism by other parties. Second, there is concern that the research results that are disclosed can easily be exploited commercially by parties who do not contribute to the research process. Third, there is concern that research results that are disseminated have the potential to be "stolen" and developed into commercial products without proper attribution to the original researchers. Finally, there is an opinion that research results that have been made public will face difficulties in the formal publication process because they can be considered duplication or even categorized as self-plagiarism by scientific journals. All of these concerns, while understandable from the perspective of individual researchers, are often unsupported by substantial evidence and tend to ignore the larger benefits of open science practices for the scientific community as a whole

Part 6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES

OF THE SCIENTIFIC PROCESS

71. 6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? (*ref.: (vi) 21*)

- ☒ YES, at the national level
- ☐ YES, at the institutional level
- ☐ YES, at both the national and institutional levels
- ☐ No

72. 6.1 cont. **If yes**, please indicate which of the recommended actions they address:

- ☒ promotion of pre-prints
- ☒ exploration of open peer-review practices
- ☒ valuing and sharing of negative scientific results and associated data
- ☒ participatory methods that value inputs from societal actors, e.g. citizen science, crowdsourcing scientific projects
- ☒ development of participatory strategies for identifying the needs of marginalized communities and highlighting socially relevant issues to be incorporated into research agendas
- ☒ promoting open innovation practices
- ☐ Other

73. 6.1. cont. **If yes**, please provide more information and links as relevant

<https://doapr.coar-repositories.org/repositories/rinarxiv/>;
<https://penerbit.brin.go.id/press/catalog/book/361> (citizen science)

Part 7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT

OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

74. 7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (ref.: (vii) 22.a)

the implementation of international scientific collaboration in the context of open science in Indonesia has not been seen in a systematic and structured manner at the institutional or national level involving various institutions in an integrated manner. Although there are several efforts that should be appreciated, most activities are still limited to initiatives from small groups spread across various institutions, such as research teams in universities, research units within national research institutions (for example BRIN), non-governmental organizations that focus on research, and media platforms that have special attention to the dissemination of scientific knowledge (such as The Conversation). These initiatives, while valuable, remain sporadic and have not been coordinated within a cohesive national framework to support open science.

75. 7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: (vii) 22.b, 22.e, 22.f)

☒ Yes

☐ No

76. 7.2 cont. **If yes**, please provide links as relevant and more information.

Indonesia has demonstrated a commitment to promoting scientific collaboration at both national and international levels, which is implemented through various programs such as Indonesian Collaborative Research and International Collaborative Research. These programs are designed to encourage cross-institutional and cross-country research collaboration, with the aim of improving the quality and impact of Indonesian research on the global stage. However, there is a gap between the implementation of these programs and UNESCO's recommendations on open science, especially in terms of the evaluation metrics used. The academic guidelines that these programs refer to still rely heavily on mainstream metrics such as number of citations and journal impact factors, an approach that runs counter to the principles outlined in the UNESCO Recommendation on Open Science, which instead encourages the use of more diverse and inclusive evaluation metrics

77. 7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science? (ref.: (vii) 22.c)

☒ Yes

☐ No

78. 7.3 cont. **If yes**, please provide links as relevant and more information.

Several meetings and discussions about open science funding have taken place, but there has been no consistent follow-up. As can be seen from the various sources above, open science requires sustainable policy support and funding to be implemented effectively. Without clear regulatory and funding mechanisms, open science practices will be difficult to develop despite their proven benefits in increasing research efficiency, collaboration and innovation.

Part 8. GENERAL CONSIDERATIONS

79. 8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

☒ Yes

☐ No

80. 8.1 cont. If yes, please provide more information.

Regulations that are heavy on mainstream metrics (citation based) that are not directly or indirectly related to the UNESCO recommendations. This is reflected in the still dominant use of citation metrics in research assessment (explicitly stated in the regulations on career assessment and research performance assessment published by the ministry), while UNESCO encourages broader open science practices such as data sharing, collaboration and transparency. As shown in various studies, excessive focus on citation metrics can hinder innovation and collaboration in research. In fact, open science has been proven to increase research efficiency, encourage more innovative solutions, and strengthen collaboration between researchers.

81. ANY OTHER COMMENTS: If you wish, you may enter any additional comments or information here. You may also contact openscience@unesco.org

should you have any queries or clarifying the response, please do not hesitate to contact us